

Problem: Consider a tube with mean radius of 25 mm made of graphite-reinforced composite with a layer $[+30_6] \equiv [30^\circ]$. The tube is under axial and torsion loads. Solve the problem with three methods: Classical Laminate Theory, Analytical Solution, and Numerical Solution. At least, compare results (wall thickness is 0.9 mm).

Solution

Note: In all methods, we apply axial force as $P = 1000$ N and $T = 50000$ N.mm

CLT Method

As mentioned in the description, we have loads N_x and N_{xy} . Therefore, we can write the loading as

$$N_x = \frac{P}{2\pi R}, \quad N_y = 0, \quad N_{xy} = \frac{T}{2\pi R^2}, \quad N_x = 0, \quad M_y = 0, \quad M_{xy} = 0$$

From CLT we have

$$\begin{Bmatrix} N_x \\ N_y \\ N_{xy} \\ M_x \\ M_y \\ M_{xy} \end{Bmatrix} = \begin{bmatrix} A_{11} & A_{12} & A_{16} & B_{11} & B_{12} & B_{16} \\ A_{12} & A_{22} & A_{26} & B_{12} & B_{22} & B_{26} \\ A_{16} & A_{26} & A_{66} & B_{16} & B_{26} & B_{66} \\ B_{11} & B_{12} & B_{16} & D_{11} & D_{12} & D_{16} \\ B_{12} & B_{22} & B_{26} & D_{12} & D_{22} & D_{26} \\ B_{16} & B_{26} & B_{66} & D_{16} & D_{26} & D_{66} \end{bmatrix} \begin{Bmatrix} \varepsilon_x^o \\ \varepsilon_y^o \\ \gamma_{xy}^o \\ \kappa_x^o \\ \kappa_y^o \\ \kappa_{xy}^o \end{Bmatrix} \quad (1)$$

which

$$A_{ij} = \sum_{k=1}^N \bar{Q}_{ijk}(z_k - z_{k-1}) \quad (2)$$

$$B_{ij} = \frac{1}{2} \sum_{k=1}^N \bar{Q}_{ijk}(z_k^2 - z_{k-1}^2) \quad (3)$$

$$D_{ij} = \frac{1}{3} \sum_{k=1}^N \bar{Q}_{ijk}(z_k^3 - z_{k-1}^3) \quad (4)$$

Because all six plies have the same orientation, there is no coupling and $B_{ij} = 0$. The matrix reduces to

$$\begin{Bmatrix} N_x \\ N_y \\ N_{xy} \end{Bmatrix} = \begin{bmatrix} A_{11} & A_{12} & A_{16} \\ A_{12} & A_{22} & A_{26} \\ A_{16} & A_{26} & A_{66} \end{bmatrix} \begin{Bmatrix} \varepsilon_x^o \\ \varepsilon_y^o \\ \gamma_{xy}^o \end{Bmatrix} \quad (5)$$

For graphite-polymer composite we have

$$\theta = 30^\circ, \quad E_1 = 155.00 \text{ GPa}, \quad E_2 = 12.10 \text{ GPa}, \quad G_{12} = 4.40 \text{ GPa}, \quad \nu_{12} = 0.248$$

In the code by $[\bar{Q}] = [T]^{-1}[Q][T_\epsilon]$ we calculate

$$\bar{Q} = \begin{bmatrix} 92.8 & 30.1 & 46.7 \\ 30.1 & 21.0 & 15.5 \\ 46.7 & 15.5 & 31.5 \end{bmatrix} \text{ GPa}$$

By considering the applied load and torsion, and substituting these values alongside $[Q]$ into the main equation, we obtain the following values for each layer

$$\begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{Bmatrix} = \begin{Bmatrix} -520.7809 \\ -243.6602 \\ 1343.0200 \end{Bmatrix} \mu, \quad \begin{Bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{Bmatrix} = \begin{Bmatrix} 7.074 \\ 0 \\ 14.15 \end{Bmatrix} \text{ MPa}$$

And by transformation,

$$\begin{Bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \gamma_{12} \end{Bmatrix} = \begin{Bmatrix} 130.0438 \\ -894.4849 \\ 911.5033 \end{Bmatrix} \mu, \quad \begin{Bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{Bmatrix} = \begin{Bmatrix} 17.56 \\ -10.48 \\ 4.011 \end{Bmatrix} \text{ MPa}$$

Direct Analytical Method

Like the previous method, we apply same loading on the material. First we need to find transformed stiffness matrix and consider following parameters.

$$\bar{r} = 25 \text{ mm}, \quad t = 0.9 \text{ mm}, \quad R_1 = 24.55 \text{ mm}, \quad R_0 = 25.45 \text{ mm}$$

$$[\bar{C}] = [\bar{S}]^{-1}$$

$$[\bar{S}] = [T_1][S][T_1]^{-1}$$

For 30° from the

By performing the transformation calculations, the transformed stiffness components are

$$\begin{aligned}
\bar{C}_{11} &= 95143 \text{ MPa} \\
\bar{C}_{22} &= \bar{C}_{33} = 24,002 \text{ MPa} \\
\bar{C}_{12} &= \bar{C}_{13} = 32,718 \text{ MPa} \\
\bar{C}_{23} &= 6820 \text{ MPa} \\
\bar{C}_{44} &= G_{23} = 3200 \text{ MPa} \\
\bar{C}_{55} &= G_{12}m^2 + G_{23}n^2 = 4100 \text{ MPa} \\
\bar{C}_{66} &= G_{12}n^2 + G_{23}m^2 = 3500 \text{ MPa} \\
\bar{C}_{16} &= (m^2 - n^2)mn(C_{11} - C_{12} - 2C_{66}) = +31,038 \text{ MPa} \\
\bar{C}_{26} &= \bar{C}_{36} = (m^2 - n^2)mn(C_{12} - C_{22} + 2C_{66}) = -233 \text{ MPa}
\end{aligned}$$

For the axisymmetric tube loaded uniformly along its length, in the coordinate system, all quantities are independent of θ . The relevant strain-displacement relations (Eq. 3.3) reduce to

$$\varepsilon_r = \frac{\partial u_r}{\partial r} \quad (6)$$

$$\varepsilon_\theta = \frac{u_r}{r} \quad (7)$$

$$\varepsilon_z = \frac{\partial u_z}{\partial z} = \bar{\varepsilon}_z = \text{const} \quad (8)$$

$$\gamma_{\theta r} = -\frac{u_\theta}{r} + \frac{\partial u_\theta}{\partial r} \quad (9)$$

$$\gamma_{z\theta} = \frac{\partial u_\theta}{\partial z} = \bar{\delta} r \quad (10)$$

$$\gamma_{zr} = \frac{\partial u_z}{\partial r} \quad (11)$$

The compatibility analysis shows that ε_z is constant through the wall, and the twist rate $\bar{\delta} = d\varphi/dz$ is likewise constant (Eq. 3.52)

$$\bar{\varepsilon}_z = \text{const}, \quad \bar{\delta} = \text{const}$$

In the off-axis tube coordinates, the constitutive relation is

$$\begin{Bmatrix} \sigma_z \\ \sigma_\theta \\ \sigma_r \\ \tau_{r\theta} \\ \tau_{rz} \\ \tau_{z\theta} \end{Bmatrix} = \begin{bmatrix} \bar{C}_{11} & \bar{C}_{12} & \bar{C}_{13} & 0 & 0 & \bar{C}_{16} \\ \bar{C}_{12} & \bar{C}_{22} & \bar{C}_{23} & 0 & 0 & \bar{C}_{26} \\ \bar{C}_{13} & \bar{C}_{23} & \bar{C}_{33} & 0 & 0 & \bar{C}_{36} \\ 0 & 0 & 0 & \bar{C}_{44} & \bar{C}_{45} & 0 \\ 0 & 0 & 0 & \bar{C}_{45} & \bar{C}_{55} & 0 \\ \bar{C}_{16} & \bar{C}_{26} & \bar{C}_{36} & 0 & 0 & \bar{C}_{66} \end{bmatrix} \begin{Bmatrix} \varepsilon_z \\ \varepsilon_\theta \\ \varepsilon_r \\ \gamma_{r\theta} \\ \gamma_{rz} \\ \gamma_{z\theta} \end{Bmatrix} \quad (12)$$

For the axisymmetric, long tube (stresses independent of z and θ), the equilibrium

equations (Eq. 3.12) reduce to

$$\frac{d\sigma_r}{dr} + \frac{\sigma_r - \sigma_\theta}{r} = 0 \quad (13)$$

$$\frac{d\tau_{r\theta}}{dr} + \frac{2\tau_{r\theta}}{r} = 0 \rightarrow \tau_{r\theta} = \frac{A}{r^2} \quad (14)$$

$$\frac{d\tau_{rz}}{dr} + \frac{\tau_{rz}}{r} = 0 \rightarrow \tau_{rz} = \frac{B}{r} \quad (15)$$

Applying the boundary conditions $\tau_{r\theta}(R_1) = \tau_{rz}(R_1) = 0$ (Eq. 3.33b, 3.33d) to Eqs. (14)–(15) gives immediately:

$$A = B = 0 \rightarrow \tau_{r\theta} = \tau_{rz} = 0 \text{ for } r \quad (16)$$

The general displacement field (Eq. 3.35, with $\Delta T = 0$) simplifies to

$$u_z = \bar{\varepsilon}_z z \quad (17)$$

$$u_\theta = \bar{\delta} r z \quad (18)$$

$$u_r = G r^\beta + H r^{-\beta} + a_1 \bar{\varepsilon}_z r + a_2 \Delta T r + a_3 \bar{\delta} r^2 \quad (19)$$

Where $\beta = \sqrt{\bar{C}_{33}/\bar{C}_{22}} = 1$ for transverse isotropy and

The corresponding strain field (Eq. 3.36),

$$\varepsilon_r = G\beta r^{\beta-1} - H\beta r^{-\beta-1} + a_1 \bar{\varepsilon}_z + a_2 \Delta T + 2a_3 \bar{\delta} r \quad (20)$$

$$\varepsilon_\theta = G r^{\beta-1} + H r^{-\beta-1} + a_1 \bar{\varepsilon}_z + a_2 \Delta T + a_3 \bar{\delta} r \quad (21)$$

$$\varepsilon_z = \bar{\varepsilon}_z \quad (22)$$

$$\gamma_{r\theta} = \gamma_{rz} = 0 \quad (23)$$

$$\gamma_{z\theta} = \bar{\delta} r \quad (24)$$

$$\gamma_{\theta z} = \bar{\delta} r \quad (25)$$

The stress components entering the two global equations (Eq. 3.41–3.42) are, with $A = B = 0$ and $\Delta T = 0$

$$\sigma_z = [\bar{C}_{11} + (\bar{C}_{12} + \bar{C}_{13}) a_1] \bar{\varepsilon}_z + [(\bar{C}_{12} + \bar{C}_{13}) a_3 + \bar{C}_{16}] \bar{\delta} r \quad (26)$$

$$\tau_{z\theta} = [\bar{C}_{16} + (\bar{C}_{26} + \bar{C}_{36}) a_1] \bar{\varepsilon}_z + [(\bar{C}_{26} + \bar{C}_{36}) a_3 + \bar{C}_{66}] \bar{\delta} r \quad (27)$$

Integrating over the thin-wall cross-section

$$F_z = 2\pi \int_{R_1}^{R_0} \sigma_z r dr, \quad T_z = 2\pi \int_{R_1}^{R_0} \tau_{z\theta} r^2 dr \quad (28)$$

This gives the 2×2 stiffness system

$$\begin{bmatrix} K_{11} & K_{12} \\ K_{12} & K_{22} \end{bmatrix} \begin{Bmatrix} \bar{\varepsilon}_z \\ \bar{\delta} \end{Bmatrix} = \begin{Bmatrix} P \\ T \end{Bmatrix} \quad (29)$$

where

$$K_{11} = 2\pi \int_{R_1}^{R_0} \sigma_z r dr, \quad K_{12} = 2\pi \int_{R_1}^{R_0} \sigma_z r^2 dr, \quad K_{22} = 2\pi \int_{R_1}^{R_0} \tau_{z\theta} r^2 dr$$

The determinant of $[K]$

$$\Delta = K_{11}K_{22} - K_{12}^2$$

Note: The negative determinant reflects the dominant coupling at 30° ; The solution remains physically valid because the coupled equations are solved simultaneously.

By Cramer's rule

$$\bar{\varepsilon}_z = \frac{P K_{22} - T K_{12}}{\Delta}, \quad \bar{\delta} = \frac{T K_{11} - P K_{12}}{\Delta}$$

The mean shear strain at $r = \bar{r}$

$$\bar{\delta} = \frac{1}{2} \left(\frac{P}{T} \right) \left(\frac{K_{11}}{K_{22}} \right)$$

Back-substituting into the constitutive relations

With $A = B = 0$, $\Delta T = 0$, and the rigid-body constants set to zero, the displacement field (Eq. 3.35) is

The surface boundary conditions are satisfied exactly (Eq. 3.33)

$$\begin{aligned}\sigma_r(R_1) &= \sigma_r(R_0) = 0 \\ \tau_{r\theta}(R_1) &= \tau_{r\theta}(R_0) = 0 \\ \tau_{rz}(R_1) &= \tau_{rz}(R_0) = 0\end{aligned}$$

The constants G and H are found from the two radial-stress conditions at $r = R_1$ and $r = R_0$. The radial stress σ_r from the constitutive relation (with $A = B = 0$, $\Delta T = 0$, $a_1 = 0$, $\beta = 1$) is

$$\sigma_r = (\bar{C}_{23} + \bar{C}_{33})G + (\bar{C}_{23} - \bar{C}_{33})Hr^{-2} + [(\bar{C}_{23} + 2\bar{C}_{33})a_3 + \bar{C}_{36}]\bar{\delta}r$$

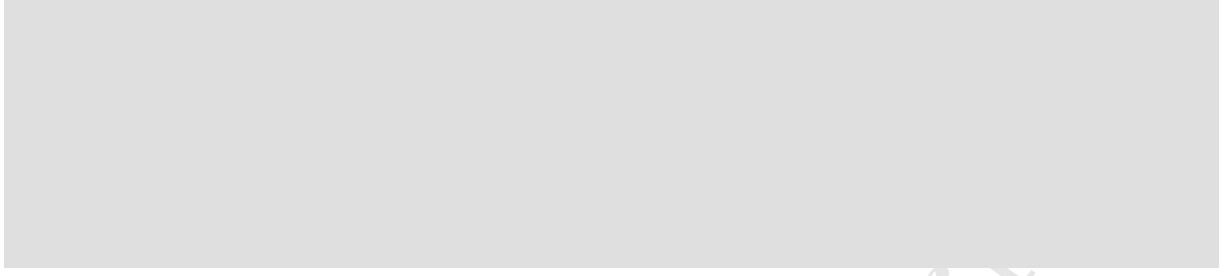
Applying $\sigma_r(R_1) = 0$ and $\sigma_r(R_0) = 0$,

$$\begin{aligned}(\bar{C}_{23} + \bar{C}_{33})G + (\bar{C}_{23} - \bar{C}_{33})HR_1^{-2} &= -[(\bar{C}_{23} + 2\bar{C}_{33})a_3 + \bar{C}_{36}]\bar{\delta}R_1 \\ (\bar{C}_{23} + \bar{C}_{33})G + (\bar{C}_{23} - \bar{C}_{33})HR_0^{-2} &= -[(\bar{C}_{23} + 2\bar{C}_{33})a_3 + \bar{C}_{36}]\bar{\delta}R_0\end{aligned}$$

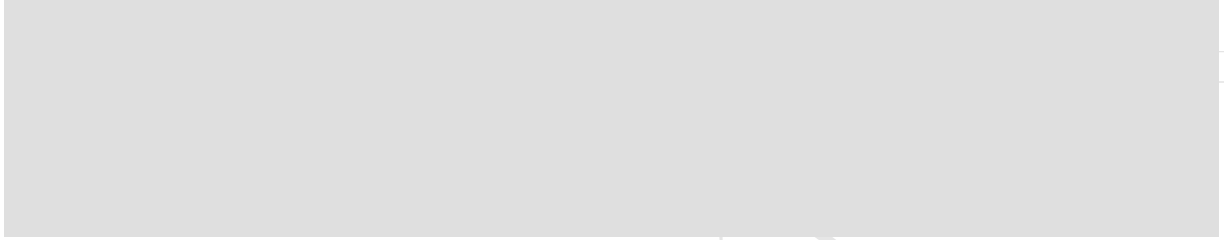
Substituting the numerical values

The right-hand sides become

Subtracting the two equations to isolate H



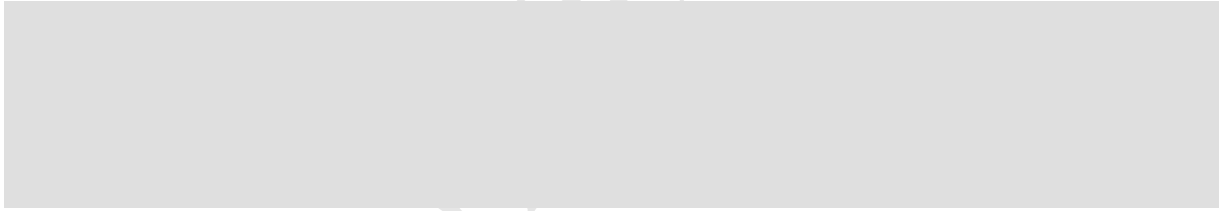
Substituting H back to find G



The circumferential strain ε_θ at any radius r is then (with $\beta = 1$, $a_1 = 0$, $\Delta T = 0$)

$$\varepsilon_\theta = G + Hr^{-2} + a_3 \bar{r}$$

Evaluating at the mean radius $\bar{r}$



$$\bar{\varepsilon}_z = 0.6572 \times 10^{-3} \text{ mm/mm}$$

$$\varepsilon_\theta = -0.0142 \times 10^{-3}$$

$$\gamma_{z\theta} = -1.7865 \times 10^{-3}$$

$$\sigma_z = 7.07 \text{ MPa}$$

$$\sigma_\theta = 0$$

$$\tau_{z\theta} = 14.147 \text{ MPa}$$

$$\begin{Bmatrix} \varepsilon_z \\ \varepsilon_\theta \\ \gamma_{z\theta} \end{Bmatrix} = \begin{Bmatrix} 657.2 \\ -14.20 \\ -1786.6 \end{Bmatrix} \mu, \quad \begin{Bmatrix} \sigma_z \\ \sigma_\theta \\ \tau_{z\theta} \end{Bmatrix} = \begin{Bmatrix} 7.074 \\ 0 \\ 14.147 \end{Bmatrix} \text{ MPa}$$

And by transformation,

$$\begin{Bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \gamma_{12} \end{Bmatrix} = \begin{Bmatrix} -704.2 \\ -332.6 \\ -2929.4 \end{Bmatrix} \mu, \quad \begin{Bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{Bmatrix} = \begin{Bmatrix} 17.556 \\ -10.482 \\ 4.011 \end{Bmatrix} \text{ MPa}$$

Numerical Method

As hinted in the question, we use ABAQUS for the numerical solution.

Part

For the model, we create a tube with shell model and extrude it 200 mm.

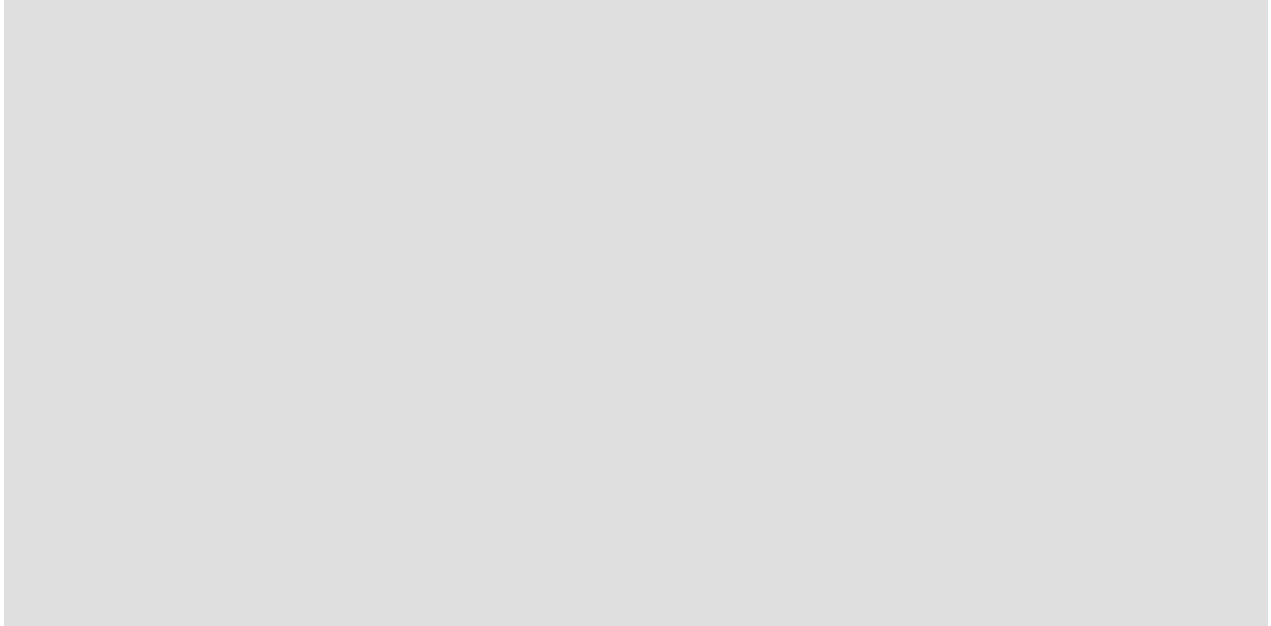


Figure 1: Cube part geometry

Property

We define the graphite-polymer properties and apply them as a lamina.

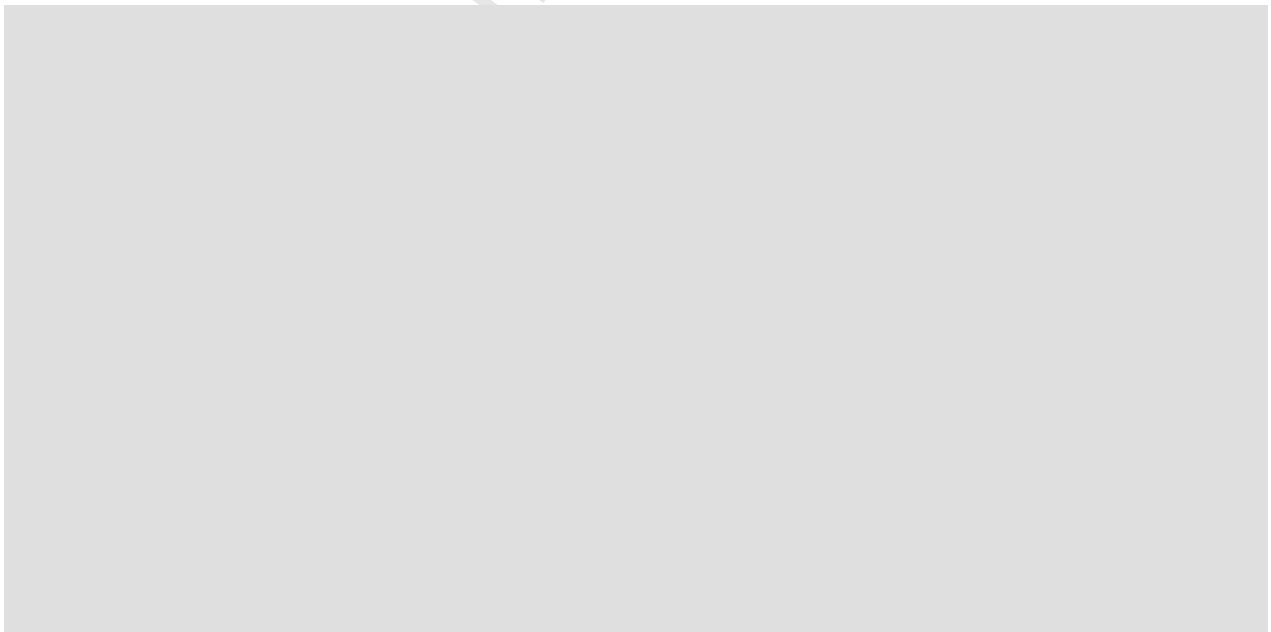


Figure 2: Material Properties

The material has a $\theta = 30^\circ$ orientation by the axial load direction. We define a datum to set a new coordinate system at the origin, which allows us to accurately define the material orientation using a cylindrical coordinate system.

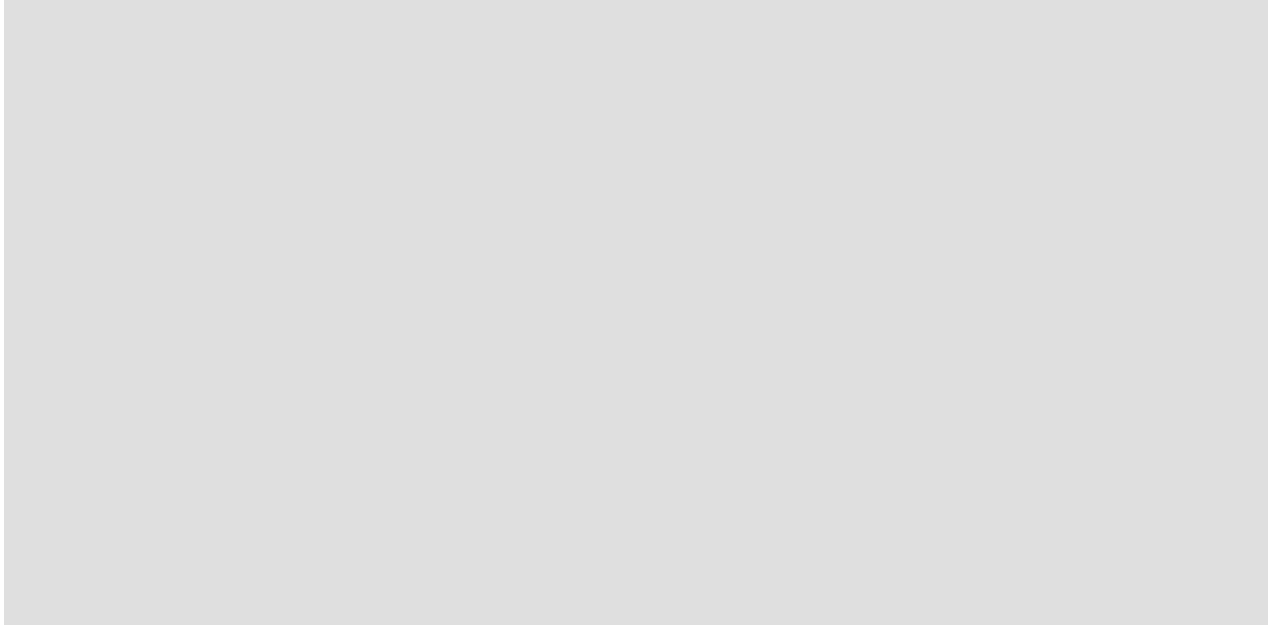


Figure 3: Create Datum

We assigned the orientation to the material. Based on the chosen directions and the axial load, to achieve a $\theta = 30^\circ$ orientation relative to the axial direction, we set the rotation angle to 60° .”

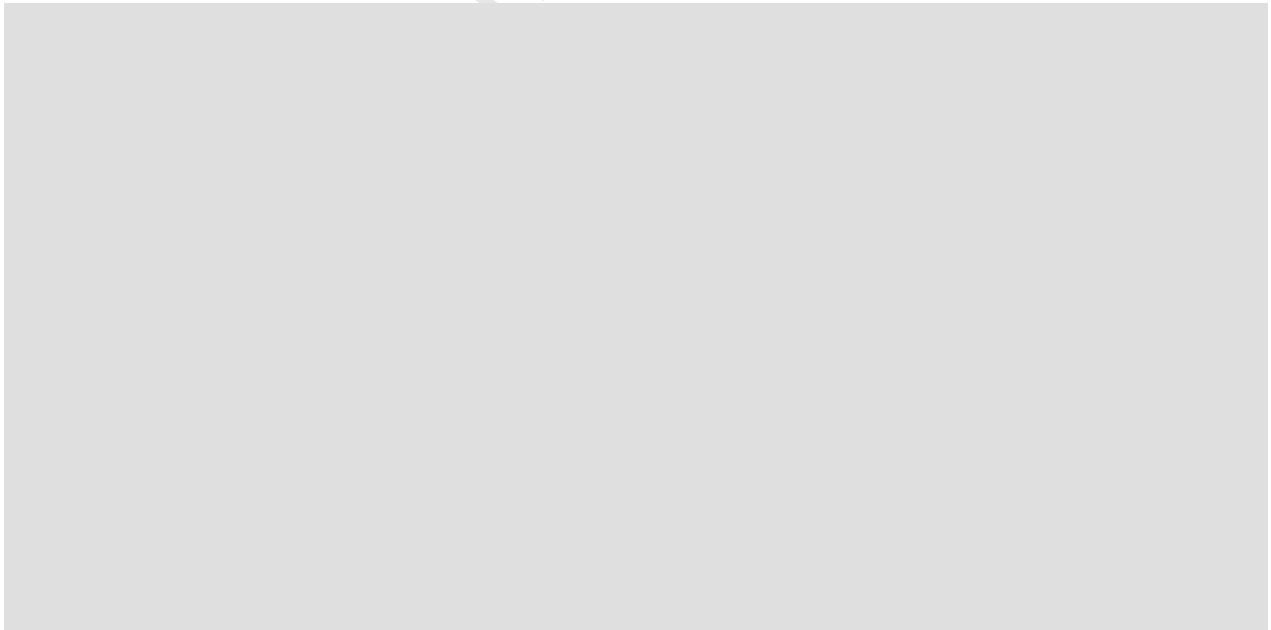


Figure 4: Assign orientation to material

Step

Create a Static, General step after the initial step. in field output, we set the domain on composite layup and set point in the middle of each layer.

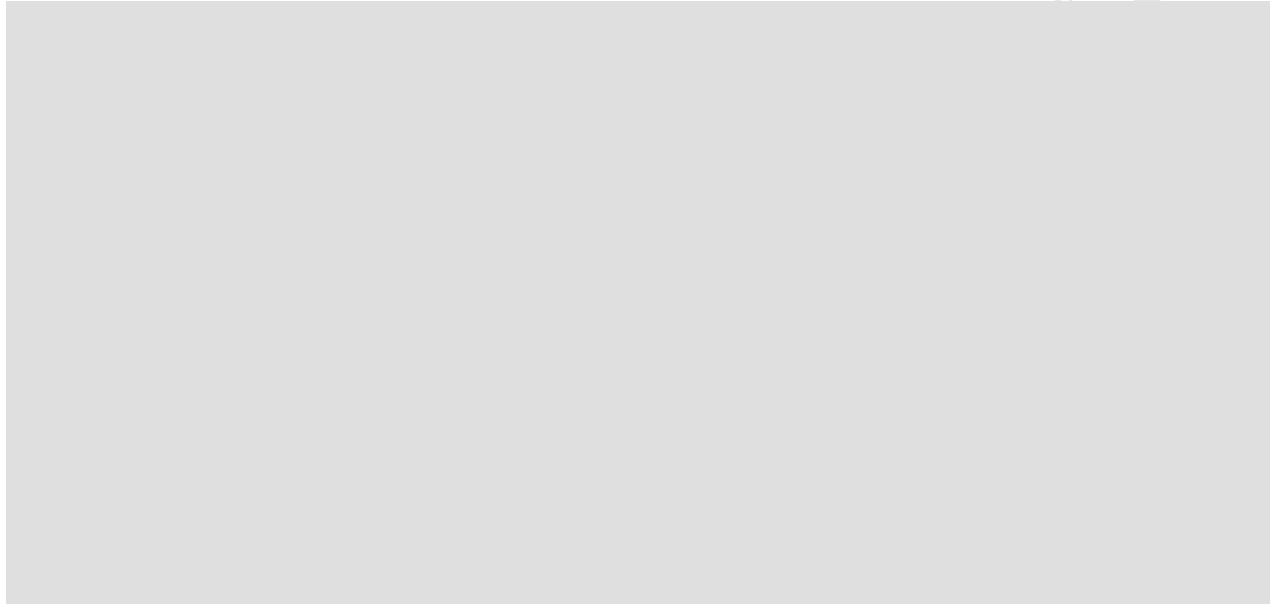


Figure 5: Creating a static, general step

Load

For loading, we applied axial load and torsion on a reference point we created with constraints.

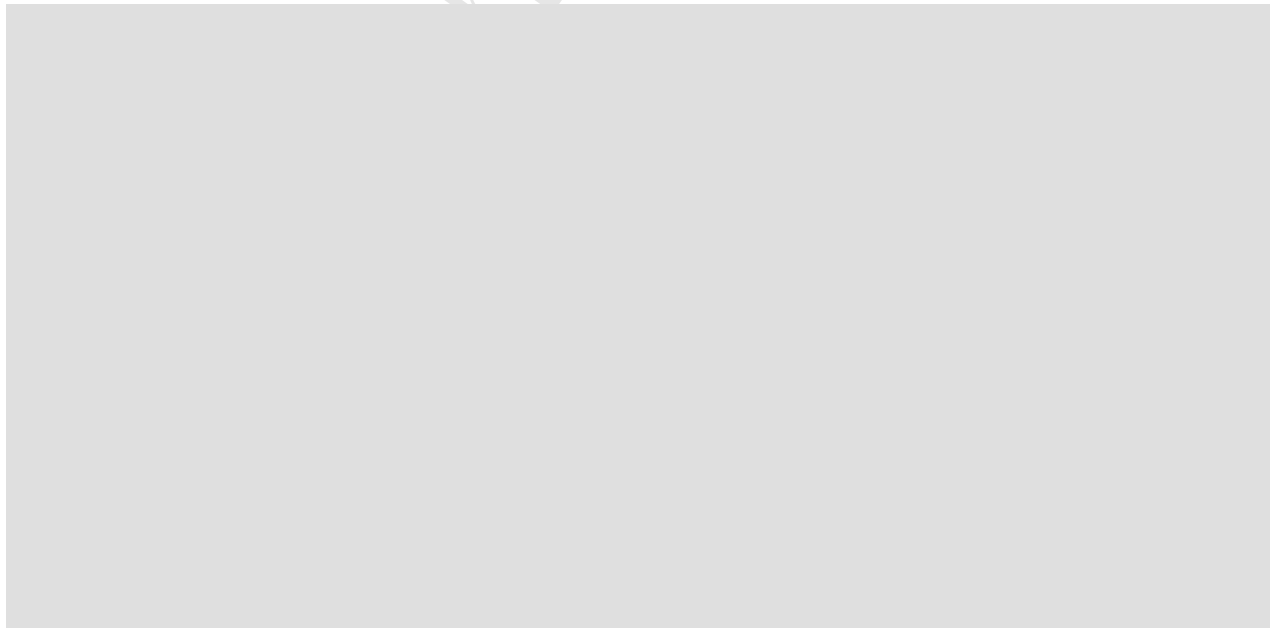


Figure 6: Defining combined axial and torsion loads in the Load module

Mesh

We used S4 elements with shell family to create the mesh.

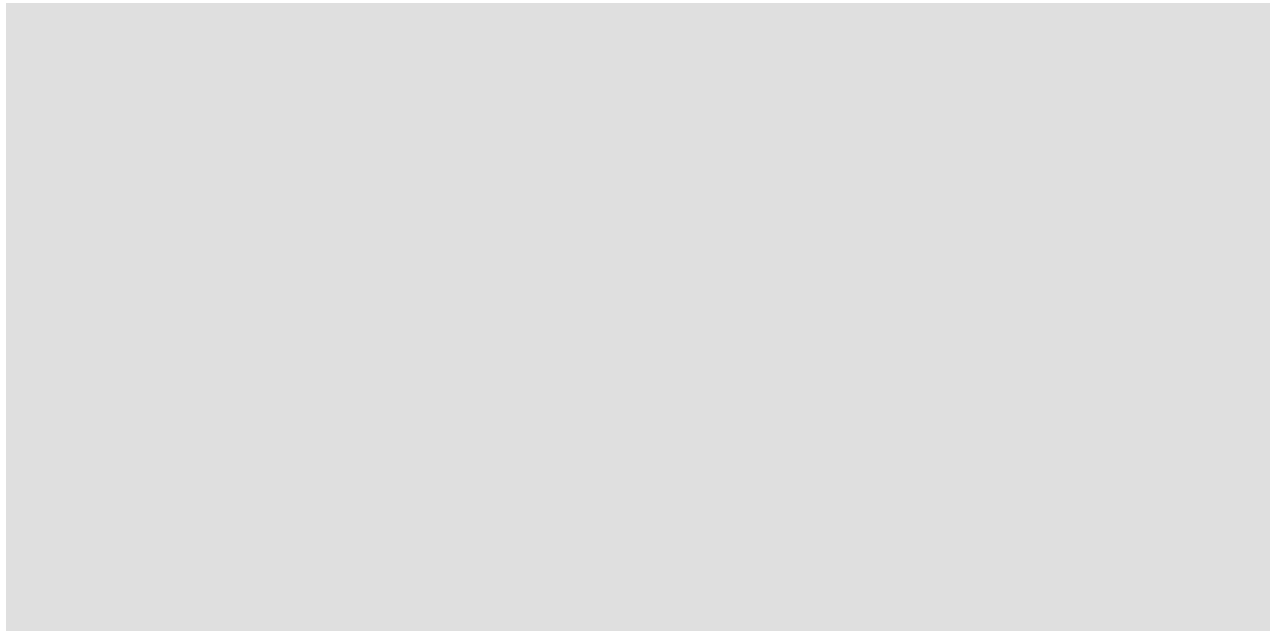


Figure 7: Single-element C3D8 mesh

Visualization

After submitting the job, we use the Query and Probe Values tools in the Visualization module. By selecting an element far from both ends of the tube, we can extract the stress and strain values.

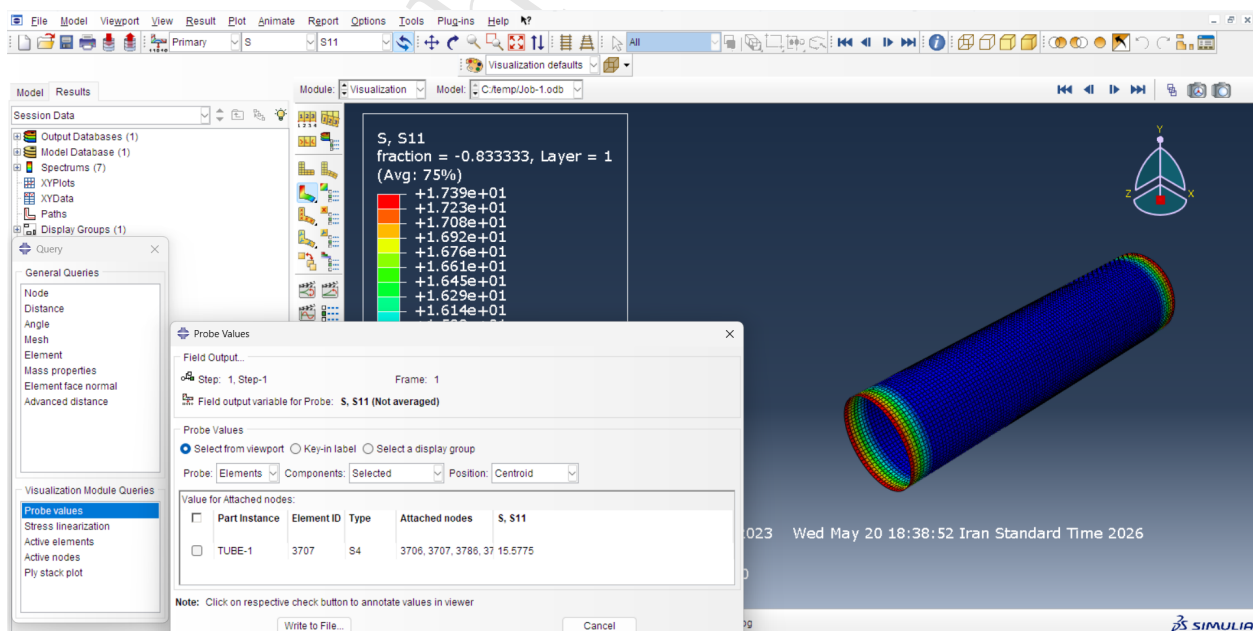


Figure 8: Taking strains as outputs

Obtained stresses and strains are as follow.

$$\begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{Bmatrix} = \begin{Bmatrix} -250.98 \\ -243.728 \\ 131.349 \end{Bmatrix} \mu, \quad \begin{Bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{Bmatrix} = \begin{Bmatrix} 5.67362 \\ -0.464 \\ 13.2079 \end{Bmatrix} \text{ MPa}$$

And by transformation,

$$\begin{Bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \gamma_{12} \end{Bmatrix} = \begin{Bmatrix} 117.089 \\ -881.797 \\ 896.850 \end{Bmatrix} \mu, \quad \begin{Bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{Bmatrix} = \begin{Bmatrix} 15.5775 \\ -10.3682 \\ 6.94614 \end{Bmatrix} \text{ MPa}$$

Table 1: Results from three methods

Symbol	CLT	Elasticity	ABAQUS
σ_z	+7.0740	+7.0740	+5.6736
σ_θ	0	0	-0.4640
$\tau_{z\theta}$	+14.1500	+14.1470	+13.2079
ε_z	-520.7809	+657.2000	-250.9800
ε_θ	-243.6602	-1694.0000	-243.7280
$\gamma_{z\theta}$	+1343.0200	-1786.6000	+131.3490
σ_1	+17.5600	+17.5560	+15.5775
σ_2	-10.4800	-10.4820	-10.3682
τ_{12}	+4.0110	+4.0110	+6.9461
ε_1	+130.0438	-704.2000	+117.0890
ε_2	-894.4849	-332.6000	-881.7970
γ_{12}	+911.5033	-2929.4000	+896.8500

The Classical Laminate Theory (CLT) and Elasticity methods demonstrate near-perfect agreement in calculating both global and principal stresses, whereas the numerical ABAQUS model predicts slightly lower overall stresses and a negligible compressive hoop stress not seen in the analytical models. In contrast to the stress results, the calculated strains vary significantly across all three methodologies. For example, global axial strain is predicted as compressive by CLT and ABAQUS, but tensile by the Elasticity method. Ultimately, these widespread discrepancies in shear and principal strains highlight how each method's unique structural and kinematic assumptions directly impact the resulting deformation fields.